



भारतीय विज्ञान शिक्षा एवं अनुसंधान संस्थान मोहाली

(मानव संसाधन विकास मंत्रालय का एक स्वायत्त संस्थान, भारत सरकार के अधीन)
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NAME:

ROLL NUMBER:

MID-SEMESTER EXAM EES301

Date: 09.10.2025

GLOBAL BIOGEOCHEMICAL CYCLES AND CLIMATE CHANGE

Total Marks: 25

Please read the questions carefully before answering and return the question paper with your name and roll number.

- 1) The densest (heaviest) ocean water has (1 Mark):
 - a. warm temperature, high salinity
 - b. warm temperature, low salinity
 - c. cold temperature, low salinity
 - d. cold temperature, high salinity
- 2) What direction does air move according to the pressure gradient force? (1 Mark)
 - a. High pressure to low pressure
 - b. Low pressure to high pressure
 - c. Left in the southern hemisphere and right in the northern hemisphere
 - d. Right in the southern hemisphere and left in the northern hemisphere
- 3) Which of the greenhouse gas in the following list is most abundant in our atmosphere? (1 Mark)
 - a) carbon dioxide CO_2
 - b) methane CH_4
 - c) nitrous oxide N_2O
 - d) SF_6
- 4) We can use isotopes of oxygen to determine the climate of the past. Cool climates have higher ratios of $\delta^{18}\text{O}$ in the shells of deep water organisms because: (1 Mark)
 - a) The heavier ^{18}O preferably evaporates from the ocean
 - b) The heavier ^{18}O is preferably incorporated into the shell
 - c) More isotopically light ice was trapped in the large continental ice sheets and the water in the oceans was heavier
 - d) The biomes changed and we study different organisms as ice age and warm age proxy
- 5) The term albedo refers to (1 Mark)
 - a) The density of air in the atmosphere
 - b) The fraction of solar radiation reflected from Earth's surface
 - c) The fraction of solar radiation absorbed by Earth's surface
 - d) The amount of ozone in the atmosphere
- 6) If the precipitation on the Indian Ocean is 101 cm yr^{-1} then how much rain has fallen? Its area is $74.92 \times 10^6 \text{ km}^2$ (1 Mark).
- 7) Strontium has a concentration of 7.8 mg/kg in the ocean and of $320 \text{ } \mu\text{g/g}$ in the upper crust. What is its estimated lifetime in the ocean. Use $\log \tau_0 = 1.01 \log K_D + 8.4$ to estimate. (1 Mark)
- 8) The concentration of methane (CH_4) has been determined as $8 \times 10^{13} \text{ molecules / cm}^3$ at 298 K , 1 atm , $R = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1}$. $N_A = 6.022 \times 10^{23}$. Express this in ppmv (2 Marks)
- 9) How many mol of H_2CO_3^* will be there in rainwater in equilibrium with 2000 ppm CO_2 (anticipated RCP8.5 year 2300 levels) (1 Mark). Also calculate the pH of rainwater containing only carbonic acid expected in response to the rise in CO_2 . $K_H = 29 \text{ atm mol}^{-1} \text{ L}$, $K_1 = 4.5 \times 10^{-7}$, $N_A = 6.022 \times 10^{23}$. (2 Marks)
- 10) Calculate the saturation water vapor pressure and actual in an air parcel with a temperature of 20°C and 60% RH using the Thetens equation: (2 Marks)
$$P = 0.61078 \exp\left(\frac{17.27T}{T + 237.3}\right)$$
- 11) The solar constant for Venus is 2640 W m^{-2} and the planetary albedo of Venus is 0.75 . What would its effective temperature be as observed from space? (Stefan Boltzmann constant $= 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. Even so, Venus is closer to the Sun than the Earth, in this case why is its effective surface temperature lower than that of the Earth ($T_{\text{eff}} = 255 \text{ K}$) (2 Marks)

- 12) What is the golden peak and how has it been used to study ocean circulation and deep water formation? (1 Mark)
- 13) Explain the difference between gross and net ocean productivity! (1 Mark)
- 14) Draw a vertical cross section of the troposphere from the equator to the pole which displays the motion of the major atmospheric circulation cells, name the three cells and indicate the location of the two jet streams and the place where clouds will form (2 Marks). Explain how the distribution of tropical rainforests, temperate forest, cold deserts and the hot arid desert belt is related to the three large atmospheric circulation cells. At which latitude North and South of the Equator will tropical rainforests, temperate forest, cold deserts and the arid desert belts typically be located? (1 Mark)
- 15) Why is the net oxygen source flux from the ocean to the atmosphere at any point in geological time limited by the organic carbon burial rates? (1 Mark)
- 16) Write the four reactions that make up the Chapman cycle of stratospheric ozone formation (1 Mark)
- 17) Write the reactions that show how chlorine catalytically destroys ozone (1 Mark). What does CFC stand for and why was CFC manufacturing and sale banned via the Montreal Protocol? (1 Mark)
- 18) Define: greenhouse gas, global warming potential, radiative forcing, short-lived climate (1 Mark)